

NAIN, NL, Canada

WMO: 716650

Lat: 56.550N

Lon: 61.683W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.2	-26.7	-35.1	0.1	-27.6	-33.1	0.2	-26.4	20.7	-14.7	18.8	-14.2	6.1	280	0.912

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	22.4	15.6	19.7	14.1	17.4	12.8	16.4	21.2	14.7	19.1	13.1	16.9	4.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	10.1	19.2	12.4	8.9	17.2	10.7	8.0	15.5	46.3	21.1	41.3	19.3	37.1	17.0	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.8	13.0	-30.6	28.4	1.9	2.5	-32.0	30.2	-33.2	31.6	-34.3	33.0	-35.7	34.8		
(5)			-31.0	19.9	1.9	1.5	-32.4	21.0	-33.5	21.9	-34.6	22.7	-36.0	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-16.4	-16.9	-11.9	-4.5	1.8	6.6	10.2	11.0	7.8	2.7	-3.6	-10.6	(6)
(7)		DBStd	10.83	6.87	6.45	6.42	4.90	3.12	3.72	3.90	3.38	3.23	3.05	3.99	6.26	(7)
(8)		HDD10.0	4487	817	754	680	433	255	117	47	27	82	228	408	639	(8)
(9)		HDD18.3	7389	1076	988	938	683	512	354	253	227	316	486	658	897	(9)
(10)		CDD10.0	144	0	0	0	0	1	14	53	59	16	1	0	0	(10)
(11)		CDD18.3	4	0	0	0	0	0	1	2	1	1	0	0	0	(11)
(12)		CDH23.3	63	0	0	0	0	0	14	29	18	2	0	0	0	(12)
(13)		CDH26.7	15	0	0	0	0	0	4	7	4	0	0	0	0	(13)
(14)	Wind	WSAvg	5.2	6.2	5.8	5.8	5.5	4.4	4.1	3.7	3.7	5.0	5.7	6.3	6.7	(14)
(15)	Precipitation	PrecAvg	905	86	62	74	72	57	73	107	76	77	77	65	69	(15)
(16)		PrecMax	1214	252	155	176	173	147	123	191	192	182	162	116	153	(16)
(17)		PrecMin	652	9	1	13	10	2	19	29	28	18	37	14	15	(17)
(18)		PrecStd	128	54	40	48	40	34	27	46	35	42	26	26	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	1.9	4.6	8.0	17.0	25.2	26.9	25.6	21.9	14.1	6.6	2.7	(19)
(20)			MCWB	1.4	1.4	2.2	4.2	10.6	16.2	16.6	16.9	15.2	9.9	4.6	2.1	(20)
(21)		2%	DB	0.6	0.1	1.8	5.7	11.8	18.8	22.2	22.0	17.9	10.6	4.1	1.3	(21)
(22)			MCWB	0.1	-0.5	0.5	2.8	7.2	12.6	15.0	15.8	13.7	7.3	2.6	0.7	(22)
(23)		5%	DB	-1.6	-3.4	0.2	3.8	8.4	15.4	19.5	19.2	15.2	8.4	2.9	0.3	(23)
(24)			MCWB	-2.2	-4.2	-0.7	1.6	5.0	10.6	14.0	14.2	11.4	5.6	1.7	-0.3	(24)
(25)		10%	DB	-5.0	-6.8	-1.9	2.1	6.2	12.5	17.0	17.0	13.1	6.6	1.7	-1.4	(25)
(26)			MCWB	-5.8	-7.7	-3.0	0.6	3.5	8.4	12.4	12.9	9.4	4.3	0.6	-2.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	1.6	2.4	4.7	10.8	17.5	18.0	18.8	16.5	10.3	5.0	2.2	(27)
(28)			MCDWB	1.8	1.9	4.1	7.5	16.9	22.9	24.4	24.5	20.3	13.3	6.2	2.7	(28)
(29)		2%	WB	0.1	-0.4	0.8	3.1	7.6	13.5	16.0	16.6	14.0	7.7	2.9	0.9	(29)
(30)			MCDWB	0.6	0.1	1.6	5.4	11.3	18.0	21.3	20.8	17.7	10.1	3.7	1.2	(30)
(31)		5%	WB	-2.2	-4.2	-0.7	1.9	5.2	10.8	14.3	14.7	11.7	5.9	1.8	-0.2	(31)
(32)			MCDWB	-1.7	-3.4	0.3	3.4	8.1	15.1	18.9	18.9	15.2	8.0	2.7	0.3	(32)
(33)		10%	WB	-5.8	-7.6	-2.9	0.9	3.7	8.6	12.7	13.1	9.7	4.7	0.7	-2.1	(33)
(34)			MCDWB	-5.0	-6.8	-2.0	2.0	5.9	12.1	16.7	16.6	12.6	6.2	1.5	-1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.5	8.3	7.5	6.6	7.7	7.9	7.9	6.8	4.8	4.9	5.6	(35)
(36)			MCDBR	7.8	8.1	9.1	9.2	11.3	13.8	13.6	12.8	10.9	7.6	5.6	4.7	(36)
(37)			MCWBR	7.9	7.8	8.5	7.6	7.9	9.0	8.4	8.3	7.7	5.8	5.2	4.8	(37)
(38)		5% WB	MCDBR	7.9	8.0	8.8	7.9	10.7	13.2	13.0	12.5	10.5	7.1	5.3	4.3	(38)
(39)			MCWBR	8.0	7.8	8.3	6.8	7.7	9.2	8.7	8.7	7.9	5.6	5.2	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.230	0.250	0.277	0.331	0.398	0.379	0.383	0.364	0.331	0.294	0.250	0.235	(40)	
(41)		taud	2.212	2.246	2.199	2.094	2.063	2.332	2.400	2.458	2.513	2.529	2.298	2.104	(41)	
(42)		Ebn at Noon	693	829	888	879	835	862	851	846	830	779	659	552	(42)	
(43)		Edh at Noon	51	75	103	135	150	117	107	95	78	59	47	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.64	1.64	3.15	4.76	5.72	5.24	4.58	3.95	2.85	1.48	0.69	0.43	(44)	
(45)		RadStd	0.09	0.20	0.36	0.43	0.54	0.56	0.43	0.38	0.21	0.11	0.07	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page